

Pairing, Stripes, and Competing Order in Coupled Hubbard Ladders

Introduction, Background, and Current Results

Ladder MPS+MF project
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Abstract

Weakly coupled two-leg Hubbard ladders provide a setting in which the formation of singlet pairs can be studied separately from the collective order selected by coupling correlated subsystems. This draft develops the motivation for a multichannel matrix-product-state plus mean-field study of superconducting, charge, and magnetic order, with particular attention to the relationship between stripe-centered pairing proposals and the broader picture of intertwined orders. It also collects the project's current local evidence. The isolated repulsive ladder exhibits a finite-system spin gap and substantial pairing correlations, whereas the latest square-array trajectories at zero nearest-neighbor interaction develop strong stripe textures. At an attractive interaction, two contrasting initial states instead approach a common paired state. The completed unfrustrated-cubic grid instead reaches stripes at all nine tested coordinates. Finer square cuts retain distinct textures near the boundary, with a small change in energy slope along the interaction cut but no resolved first-order singularity. Three completed repulsive- V square starts lose pairing; the first completed trellis stripe start develops it. These observations motivate a distinction between local pairing tendencies and the order selected in an array. The numerical results remain finite-length, finite-bond-dimension evidence, with explicit limits on self-consistency, response completeness, and thermodynamic interpretation.

Draft scope. Sections 1–2 are draft manuscript prose. Section 3 is a dated collection of results and interpretation, including unresolved points that must remain visible during manuscript development. The evidence cutoff is the synchronized analyses through 18 September 2026; the literature discussion also incorporates the 15 September synthesis. Appendix A identifies the local sources. No new many-body calculation is reported here.

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1 Introduction

The relationship between superconductivity, antiferromagnetism, and spatial charge organization is a central question in doped Mott insulators. The Hubbard model provides a minimal language for studying how kinetic energy and local repulsion generate collective behavior, while its extensions allow the influence of nonlocal interactions and lattice structure to be examined systematically [1–4]. In this setting, strong singlet correlations need not lead directly to superconductivity. Charge and spin order can develop from the same interacting electrons, and the scales associated with pair formation and collective phase coherence can differ. A microscopic account of superconductivity must therefore address the competing ways in which a correlated electronic state can organize.

Stripe order makes this issue particularly concrete. Experiments in lanthanum-based cuprates established charge modulations accompanied by incommensurate magnetic order, with antiferromagnetic domains separated by hole-rich regions [5]. Subsequent measurements found closely related magnetic excitation spectra in stripe-ordered and superconducting materials, motivating theories in which stripe correlations participate in the pairing mechanism [6]. A complementary view regards stripes as a competing organization of a doped Mott insulator whose short-distance singlet physics can also support superconductivity without requiring a particular stripe arrangement [3]. The resulting question concerns both the origin of pairing and the consequences of charge organization for the coherence of pairs.

The distinction between these views is sharpened by separating the existence of stripe correlations from their necessity for superconductivity. In stripe-ordered $\text{La}_{1.875}\text{Ba}_{0.125}\text{CuO}_4$, transport and magnetization measurements reveal superconducting correlations within the planes despite strong suppression of three-dimensional phase coherence [7]. A spatially sign-changing superconducting order parameter can frustrate interplane Josephson coupling and offers a specific explanation of this separation [8]. Such observations permit cooperation at the level of local pairing and competition at the level of collective order. They motivate treating superconducting, charge, and spin degrees of freedom together, as emphasized in the framework of intertwined orders [9, 10].

Modern Hubbard-model calculations reinforce the dependence of this relationship on the microscopic Hamiltonian. Complementary numerical methods find stripe order with small energy differences between wavelengths in a studied regime of the nearest-neighbor square-lattice model [11]. Other calculations find no superconducting ground state in specified regimes of that minimal model [12], while additional hopping terms can support pairing together with spatial charge modulation [13, 14]. Deliberately imposed hopping inhomogeneity can also enhance superconducting correlations on cylinders [15]. These results do not give a universal sign to the influence of stripes. They instead identify a competition that must be resolved at the level of a specified model, geometry, and numerical accuracy.

Two-leg ladders provide useful correlated building blocks for this problem. Their magnetic spectrum differs qualitatively from that of a single chain, and doped repulsive ladders can support a spin gap, bound holes, and slowly decaying singlet-pair correlations [16–18]. The corresponding low-energy description can contain a gapped spin sector and a gapless total-charge mode [19, 20]. High-accuracy numerical studies show that the relative decay of pair and density correlations requires careful control of open boundaries, bond dimension, and fit windows [21, 22]. Thus a ladder can exhibit substantial pairing tendencies while retaining the fluctuations characteristic of an effectively one-dimensional system.

Coupling such ladders introduces a second question: which order develops when the subsystems reinforce one another? Stripe-centered theories provide an explicit route from spin-gap formation to pairing and then to Josephson coherence [23]. A spatially modulated repulsive Hubbard construc-

tion demonstrates how nearly decoupled ladders can supply a pairing scale while weak interladder coupling establishes superconducting order [24]. Matrix-product-state plus mean-field methods provide a practical numerical realization of the broader coupled-subsystem strategy [25, 26]. Their multichannel extension shows why charge and superconducting solutions must be retained and compared within a common framework [27].

The interaction between neighboring densities supplies an additional tuning parameter. Calculations on extended Hubbard ladders and cylinders find that attraction can enhance pairing over selected parameter ranges, while strong attraction can also favor charge segregation [28, 29]. Recent cuprate-ladder spectroscopy, interpreted together with DMRG, provides experimental motivation for considering attraction beyond the simple onsite Hubbard interaction [30]. Its implications for a coupled array remain to be determined: stronger binding changes both the properties of the individual ladder and the effective interladder couplings. Pair mobility, susceptibility, and competing order must be considered along with binding.

The present project investigates these issues in arrays of repulsive two-leg extended Hubbard ladders, treating the ladder with an MPS ground-state method and the transverse environment with a multichannel mean-field construction. The principal regime is $U/t = 8$ and density $n = 15/16$ per physical site. Rung hopping, nearest-neighbor interaction, and transverse connectivity are varied while pairing, normal exchange, and spin-resolved density fields remain available. The immediate objective is to establish reproducible ordered solutions and understand how their competition relates to the underlying ladder correlations. The current evidence suggests that intrinsic pairing tendencies can coexist with an array tendency toward static stripes, and that attraction can reverse the observed outcome at a representative square-array point. Establishing the extent and thermodynamic stability of these behaviors requires the numerical controls discussed below.

2 Physical background and connection to the model

2.1 Local singlets, pairing correlations, and coherence

Three observables answer different questions. A spin gap measures the energy cost of the lowest accessible spin excitation at fixed particle number. A pair-binding energy compares even- and odd-particle sectors. A pair correlation function measures how a pair created in one region correlates with a pair removed elsewhere. None of these quantities alone establishes a nonzero superfluid stiffness or a superconducting transition temperature. For example, an isolated number-conserving ladder has zero anomalous expectation value even when its pair correlations decay algebraically.

For a spin-singlet bond operator,

$$\Delta_{ab} = \frac{c_{a\uparrow}c_{b\downarrow} - c_{a\downarrow}c_{b\uparrow}}{\sqrt{2}}, \quad P_{ab,cd} = \langle \Delta_{ab}^\dagger \Delta_{cd} \rangle, \quad (1)$$

the relative signs between different bond orientations characterize the internal pair structure. Opposite signs on leg and rung bonds are commonly described as *d*-wave-like in a two-leg ladder. The reduced lattice symmetry and discrete transverse momentum should be kept in mind when connecting this terminology to the gap of a two-dimensional cuprate.

An order parameter with a nonzero center-of-mass wavevector is a different property. Schematically,

$$\langle \Delta_b(R) \rangle = f_b [\Psi_0 + \Psi_Q \cos(QR + \phi)], \quad (2)$$

separates the internal bond form factor f_b from the spatial structure of the condensate. An internal leg/rung sign change alone is not evidence of a pair-density wave. Nor does a boundary-induced

modulation of a predominantly uniform pair field establish a distinct finite-momentum superconducting phase [10].

2.2 Two views of stripes and their modern interpretation

In a common-Mott-physics interpretation, singlet and magnetic correlations provide a shared microscopic starting point for several possible ordered states. Static stripes are then one possible competitor, and their presence need not be a prerequisite for pairing [3]. In a stripe-centered interpretation, spatial organization helps create the conditions for pairing. The spin-gap proximity mechanism, for example, couples a conducting stripe to a magnetically correlated environment; pair hopping transmits a spin gap to the mobile carriers, while Josephson coupling acts at a separate stage to establish coherence [23].

These interpretations share an emphasis on strong correlations and singlets. Their sharper disagreement concerns whether stripe organization is an essential intermediate step. The observations of planar superconductivity in stripe-ordered LBCO and the associated layer-decoupling proposal show how static stripes can coexist with pairing while suppressing a particular route to bulk coherence [7, 8]. Conversely, coexistence in a model or material does not establish that stripes caused its pairing.

The relation to the pseudogap remains unresolved. NMR measurements in Eu-LSCO find spin-stripe order extending practically to the pseudogap endpoint and identify competition with superconductivity, providing evidence for a close connection without determining its causal origin [31]. The spin gap of an isolated ladder, an electronic pseudogap in a cuprate, and the superconducting quasiparticle gap are different observables. A calculation of one should not be presented as a measurement or explanation of the others.

2.3 What magnetic spectroscopy establishes

Incommensurate magnetic peaks can be understood through antiphase spin domains. With lattice spacing set to one, a schematic stripe texture is

$$\langle S_{x,y}^z \rangle = m(-1)^{x+y} \cos(\kappa x), \quad q_c = 2\kappa, \quad \mathbf{q}_s = (\pi \pm \kappa, \pi). \quad (3)$$

The charge period is half the period of the staggered spin envelope. This spatial relation is useful evidence for an antiphase stripe texture, but its presence on a finite open ladder does not determine the thermodynamic amplitude or preferred wavelength.

The similarity between magnetic spectra in stripe-ordered LBCO and YBCO was an important motivation for stripe-centered interpretations [6]. Quantum stripe calculations show that low-energy spin-wave-like excitations and higher-energy ladder-like triplet excitations can occur within the same ordered model [32]. Consequently, a failure of a simple semiclassical intensity pattern does not rule out stripes or all low-energy spin waves. The connection between a spectral shape and a microscopic pairing mechanism requires more than qualitative resemblance.

The magnetic ladders in such spin-only constructions represent magnetic regions of a stripe pattern. The present model instead uses doped electronic ladders containing mobile charge. Moreover, its current analyses concern ground-state profiles and equal-time correlations, rather than a computed $S(\mathbf{q}, \omega)$. They can test the static antiphase relation in Eq. (3), but cannot yet demonstrate an hourglass spectrum or identify a neutron resonance.

2.4 Extended Hubbard ladders and transverse coupling

Each ladder has L rungs, two legs $\ell = 0, 1$, and $2L$ physical sites. Its Hamiltonian is

$$H_{\text{lad}} = -t \sum_{x=1}^{L-1} \sum_{\ell, \sigma} (c_{x\ell\sigma}^\dagger c_{x+1, \ell\sigma} + \text{H.c.}) - t_0 \sum_{x, \sigma} (c_{x0\sigma}^\dagger c_{x1\sigma} + \text{H.c.}) \\ + U \sum_{x, \ell} n_{x\ell\uparrow} n_{x\ell\downarrow} + V \sum_{\langle a, b \rangle_{\text{lad}}} n_a n_b. \quad (4)$$

The final sum includes nearest-neighbor leg and rung bonds. Thus $V < 0$ is attractive in the density channel. Open boundaries are used along the ladder; $n = N/(2L)$ is the density per physical site and $p_h = 1 - n$ the hole density. The rung hopping t_0 and interladder hopping $t_\perp$ must not be confused with the next-nearest-neighbor hopping t' in two-dimensional Hubbard studies.

The transverse reduction uses the positive scale

$$g = \frac{t_\perp^2}{|E_p|}, \quad E_p^{(h)} = E(N-2, S_z=0) + E(N, S_z=0) - 2E(N-1, S_z=\frac{1}{2}), \quad (5)$$

where negative $E_p^{(h)}$ denotes favorable hole binding. Geometry-dependent kernels convert anomalous, exchange, and density correlators into fields for the next ladder solve. The density is targeted through a chemical potential. The present ansatz retains real, spin-diagonal normal fields and real anomalous fields; it does not provide unrestricted complex phases or noncollinear magnetism (Evidence 1).

For illustration, define $\gamma = 2g$. The same-leg pairing kernels, organized by the two leg indices, are

$$K_\alpha^{\text{square}} = \gamma \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad K_\alpha^{\text{cubic, u}} = \gamma \begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix}, \quad K_\alpha^{\text{cubic, f}} = \gamma \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \quad (6)$$

These matrices summarize one part of the full map, not the entire Hamiltonian. An equation-level comparison identifies the repulsive-ladder benchmark of Ref. [26] with the branch called *cubic frustrated* in this project. That name is the project's nomenclature. Its published result should not be assigned to all three geometries (Evidence 2).

This construction is conceptually related to coupled-ladder pairing mechanisms but does not implement every assumption of the original stripe proposals. In particular, the explicit alternating dopings in the period-four construction of Ref. [24] differ from an array of equivalent ladders, although that work also considers an equivalent-ladder period-two limit.

We study equivalent coupled ladders as a prescribed microscopic geometry. The observed stripe texture is an additional longitudinal charge and spin modulation; within the repeated-ladder embedding, its domain walls extend transverse to the ladder axes. This transverse alignment is a restriction of the embedding rather than an independently optimized arrangement of distinct ladders. Collective transverse quantum fluctuations are approximated by static mean fields.

3 Results and interpretation accumulated to date

3.1 Evidence hierarchy and common parameters

The strongest current comparison concerns the square geometry at $t_0/t = 1.4$, $t_\perp/t = 0.1$, $U/t = 8$, and $n = 15/16$, with $V/t = 0$ and -0.4 . The latest paired and striped starts use $L = 64$ rungs

and bond dimension $\chi = 200$. A separate $V = 0$ comparison tests $\chi = 400$, and the isolated-ladder backbone reaches $\chi = 1200$. These are different calculations with different error controls, rather than one uniformly converged data set.

It is useful to distinguish an observed texture, a numerically self-consistent state, stability against specified perturbations, and a thermodynamic ground state. The recent trajectories provide strong texture and basin information. Their remaining acceptance failures and the incomplete length and bond-dimension studies prevent the last two categories from being identified without qualification. Here, *basin* describes access under a specified numerical map, not a measured free-energy basin.

3.2 The isolated ladder already has substantial pairing tendencies

The six-sector backbone at $V = 0$ and $t_0/t = 1.4$ gives the finite-system quantities in Table 1. All six final $\chi = 1200$ sectors pass their stored numerical gates. Changes from $\chi = 800$ are useful sensitivity observations, but not all sectors pass at that earlier stage; the comparison therefore does not constitute a complete bond-dimension extrapolation (Evidence 3, Evidence 5).

Table 1: Isolated-ladder evidence at $L = 64$, $U/t = 8$, $n = 15/16$, $t_0/t = 1.4$, and $V = 0$, from the final $\chi = 1200$ backbone. Negative binding energies use Eq. (5)’s convention.

Quantity	Value	Interpretation
Spin gap Δ_s/t	0.1533613	Fixed-particle-number spin excitation
Even-particle charge gap $\Delta_{c,2}/t$	0.01085830	Finite-size charge curvature
Hole binding $E_p^{(h)}/t$	-0.1465102	Favorable two-hole binding
Particle binding $E_p^{(e)}/t$	-0.1485427	Favorable two-particle binding
Pair decay exponent	0.7614 ± 0.0215	Fit to rung-pair correlations
Charge decay exponent	1.2158 ± 0.0830	Less reliable fit to charge correlations

The measured pair correlations decay more slowly than the fitted charge correlations. Pair fits have R^2 between 0.953 and 0.982, while charge fits give 0.527–0.707. The quoted fit errors do not include the full finite-size, boundary, and truncation uncertainty. These results are consistent with a spin-gapped ladder with appreciable pairing correlations, but do not by themselves establish its asymptotic exponents or the ordered state of an array [21, 22].

Two other diagnostics illustrate the importance of conventions and fit quality. The stored charge parameter is 1.4247 with rung normalization and 0.7124 with site normalization. An unqualified symbol K_ρ would therefore be ambiguous. A central-charge fit yields approximately 2.009 with $R^2 = 0.155$ and is excluded from physical interpretation. The open-boundary density profile has a strong component at $q/\pi = 1/8$, which can arise as a Friedel response and is not independent proof of spontaneous charge order [33].

The Stage 1 equal-time covariance spectra are broad, with participation ranks around 48–63. The first bulk spin-odd candidate follows two edge-localized modes, and a charge mode at $q/\pi = 1/8$ first appears at rank 11. This motivated combining physically chosen fields with covariance-based directions instead of assuming that a handful of leading covariance modes would exhaust the relevant response. Equal-time covariance is itself distinct from a static susceptibility, which also depends on excitation energies.

3.3 Response discovery: a strong pairing direction with unresolved controls

The Stage 2 pilot reduces fourteen named fields to twelve independent probe directions: nine normal and three pairing directions. Its projected mean-field response has a leading square pairing multiplier of about 4.6688 and a leading positive normal multiplier of 1.1879. The cubic cases both give a leading pairing multiplier of 24.4730, while their leading normal behavior differs: the largest-magnitude normal multiplier is approximately -1.1879 for the frustrated construction and $+3.5637$ for the unfrustrated one (Evidence 4).

These values show that the isolated ladder supplies strong candidate pairing response and that transverse geometry changes the competing normal channels. They do not establish which nonlinear state wins. The discovery reciprocity error is approximately 0.121% and the normal/pair cross-block ratio 0.332%, but the maximum omitted-response norm fraction is 0.83267. Small errors within the projected matrix can therefore coexist with a large response outside the chosen subspace. The second-amplitude validation was not present in the reviewed evidence (Evidence 5).

More fundamentally, the zero-field isolated ladder is not a fixed point of the full normal mean-field map: its square-map image has a normalized field norm of approximately 0.0306655, dominated by normal fields. A derivative at that reference is a bare-ladder diagnostic, not the stability matrix of a self-consistent dressed normal state. The associated square estimate $t_{\perp}/t \simeq 0.04628$ must consequently remain a projected bare-reference onset estimate, not a phase boundary.

This resolves the apparent tension between a large bare pairing response and the later nonlinear stripe trajectories. They concern different reference states, finite-amplitude feedback, and restricted response information. Positive or negative multipliers describe growth or alternation of the specified iteration. Neither is a physical excitation frequency or a global energy criterion. A physical two-sublattice interpretation of a raw two-cycle also requires a mapping to the transverse lattice and the appropriate partner-field energy [27].

3.4 Why the earlier paired-state interpretation was revised

The loose six-seed $V = 0$ campaign initially produced six software-accepted, pairing-bearing endpoints, with a corrected energy spread of only $6.0274 \times 10^{-7}t$ per physical site. This suggested a common paired region, but the later history analysis showed that it did not establish a unique stable paired basin. Three histories stopped after six evaluations; three others showed coherent spin-density growth during raw iteration and subsequent suppression under Anderson acceleration. The conspicuous change in one trajectory is reproduced by its first Anderson coefficients, $+22.6811$ and -21.6811 (Evidence 7).

This finding does not imply that Anderson acceleration changes the fixed-point equation or that every accelerated solution is unphysical. It shows that short histories and accelerated access do not test stability against a resolved competing-order perturbation. The subsequent 95%/5% two-family experiment addresses that question by using representative paired and striped reference fields, adding a finite component of the competing order, and retaining raw updates throughout.

A related warning comes from the square $(t_0/t, V/t) = (1.0, -0.4)$ coverage point. Its *diverging* outcome is a convergence classification: the fields remain bounded while pairing gives way to a stripe-like texture and an accelerated residual spike triggers the stop. It is not evidence of a divergent physical observable or an accepted phase (Evidence 8).

3.5 Attraction: two contrasting starts approach a common paired state

At $(t_0/t, V/t) = (1.4, -0.4)$, both original 80-evaluation trajectories approach the same pairing-dominated state. The initially strong stripe’s bulk leg-odd spin Hartree RMS decreases from about $1.99 \times 10^{-2}t$ to $3.88 \times 10^{-8}t$; the pairing start ends near $2.49 \times 10^{-8}t$. Both pairing-field RMS values settle near $2.8236 \times 10^{-3}t$. Bulk RMS here uses rungs 6–59 (Evidence 9).

The physical symmetrized anomalous correlators have bulk leg amplitudes near $+0.035217$ and rung amplitudes near -0.053242 , establishing a d -wave-like internal sign pattern in the stored convention. These are not normalized versions of Eq. (1) or pair–pair correlators. The full anomalous matrices agree to approximately 5.85×10^{-5} in relative L2 norm, and the largest site-density difference is below 6.23×10^{-6} . The residual boundary density profile remains visible; its survival does not indicate a surviving strong static spin stripe.

The terminal corrected energies differ by only $7.49 \times 10^{-9}t$ per site, supporting agreement of the two trajectories at their resolved accuracy. Both historical acceptance flags remain false. The principal issues are residual fluctuations in nearly vanishing channels, an oversensitive scalar slow-mode check, and a marginal energy-window miss in one lineage. The later noise-aware history checks support the plateau interpretation, but do not retroactively certify every original gate. Thus the result is strong evidence for a common paired attractor at these numerical settings, without proof of a global minimum or higher- χ stability.

3.6 Zero attraction: stripe order grows while anomalous pairing recedes

The corresponding $V = 0$ trajectories behave differently. The stripe start reaches 80 evaluations with bulk leg-pairing field RMS $1.75 \times 10^{-10}t$ and spin Hartree RMS $0.02414t$. The pairing start ends at the wall-time limit after 62 evaluations: its pairing RMS has fallen from $0.004502t$ to $0.0007090t$, while spin RMS has grown from $0.001816t$ to $0.02251t$. Pairing falls by another 65.6% across the final five records. The trend is already present before the deadline-limited final density solve (Evidence 10).

The residual anomalous correlators in the pairing-start trajectory still have opposite leg/rung signs, but their amplitudes are decreasing rapidly. They do not establish settled coexistence. The two spin textures are similar but not identical: the bulk physical spin profiles differ by 26.4% in relative L2 norm, and the density difference can reach 0.04365 per site. The terminal global relative residuals are about 5.88×10^{-4} and 2.65×10^{-2} for the stripe and pairing starts, respectively, against a 10^{-4} threshold.

The working interpretation is that a robust stripe texture has been identified and that the finite competing-order perturbation drives the pairing-dominated start toward it under raw iteration. Strict spatial self-consistency remains pending. In particular, the stripe amplitude’s apparent saturation does not eliminate slow rearrangement of the profile. These observations support strong competition between static spin/charge order and the anomalous pairing solution at this point. They do not prove that the final stripe is the global minimum or that all local pairing correlations have disappeared.

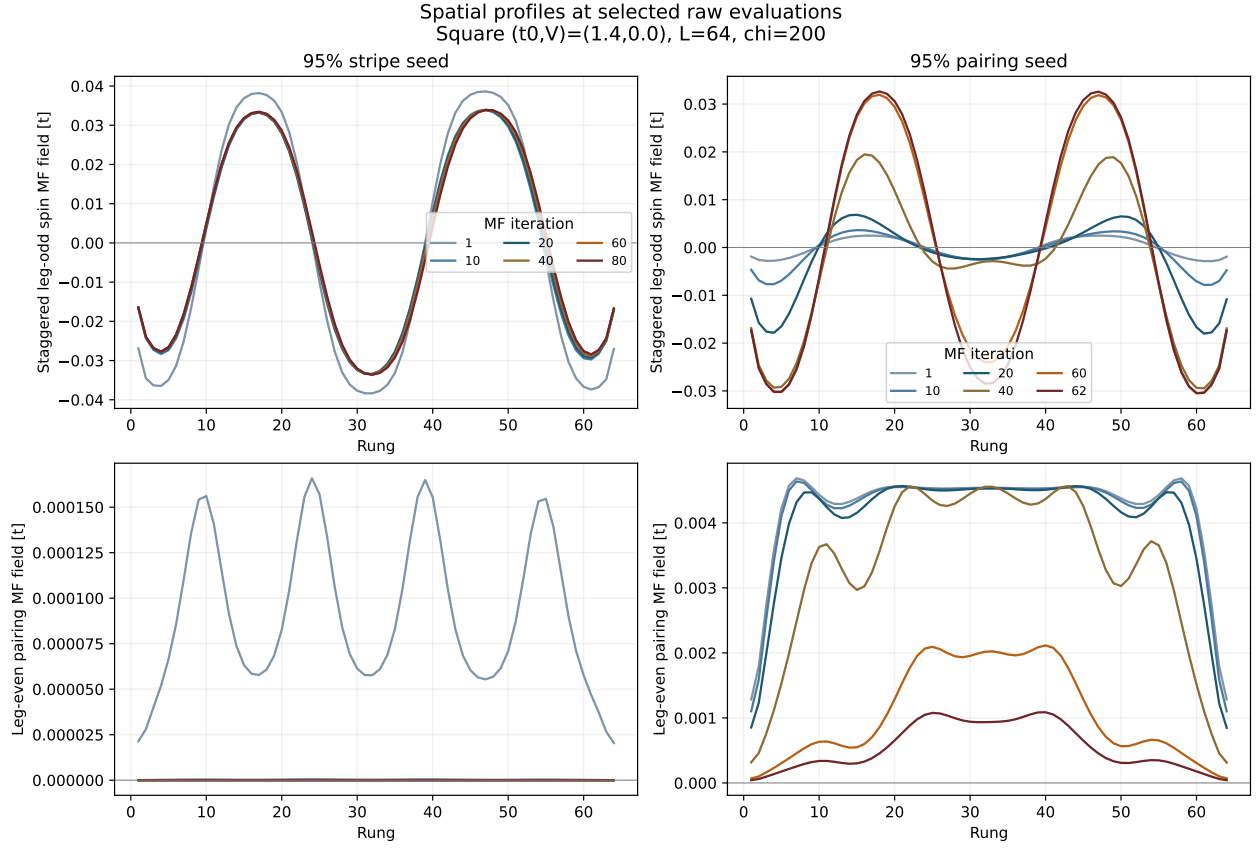


Figure 1: Previously generated spatial-history figure for the two $V = 0$ anchor trajectories. The upper row shows the staggered leg-odd spin mean field; the lower row shows the leg-even pairing mean field. The two lower panels use different vertical scales. The stripe start loses pairing, while the pairing start develops a stripe texture and its pairing decreases. Curves are self-consistency evaluations, not physical time evolution. The final states remain unaccepted for resolved field drift. Source: [Evidence 10](#).

3.7 Stripe wavevectors and slow texture relaxation

Both $V = 0$ endpoint profiles have dominant full-length Fourier modes $m_c = 4$ for charge and $m_s = 30$ for leg-odd spin. With $q_m = 2\pi m/L$ and $L = 64$,

$$q_c = \frac{\pi}{8}, \quad q_s = \frac{15\pi}{16}, \quad \kappa = \pi - q_s = \frac{\pi}{16}. \quad (7)$$

The observed relation $q_c = 2\kappa$ and the sign reversals of the staggered spin envelope near charge-depleted regions support an antiphase stripe interpretation. The charge period is 16 rungs and the spin-envelope period 32 rungs.

At hole density $p_h = 1/16$, the magnetic displacement in reciprocal-lattice units is $\delta_s = \kappa/(2\pi) = 1/32 = p_h/2$. It differs from the often discussed cuprate relation $\delta_s \simeq p_h$. In an ideal domain-wall counting picture with stripe filling ν holes per unit transverse length, $p_h = \nu/d$ and $\delta_s = 1/(2d)$; the present wavevectors would correspond to $\nu = 1$, rather than the $\nu = 1/2$ used in the usual half-filled-stripe cartoon. This is a conditional interpretation of the finite-system wavelength, not a measurement of wall filling. Neither the preferred wavelength nor its scaling with doping has been established.

The September 13 follow-up explains why the stripe looks settled while failing strict convergence. Its bulk spin RMS changes by only 0.0528% over the last twenty records, but more than 99.999% of the squared final spin residual is orthogonal to an overall rescaling of the applied spin field. Approximately 99% of that residual power lies in the bulk. The dominant remaining error concerns spatial shape.

For example, interpolated crossings of the staggered spin envelope near the third and fourth walls shift from 39.3999 and 55.1830 at evaluation 60 to 39.5165 and 55.2673 at evaluation 80. Unequal shifts reveal texture rearrangement. They are sub-rung profile diagnostics, not measured motion of physical objects. The stopping diagnostic estimates a contraction near 0.99059, suggesting very slow relaxation if that behavior persists; the extrapolation is neither a certified error bound nor a physical soft-mode eigenvalue (Evidence 10).

3.8 Energy comparisons and what the bond-dimension check adds

The relevant energy is the canonical variational functional, including the mean-field double-counting contribution. If L_{lin} denotes the expectation of the applied linear mean-field terms,

$$E_{\text{var}} = E_{\text{eff}} + \mu N + (E_{\perp} - L_{\text{lin}}) = \langle H_{\text{lad}} \rangle + E_{\perp}. \quad (8)$$

The density-targeted diagnostic is $E_{\star} = E_{\text{var}} + \mu(N_{\star} - N)$, a leading correction to the target density. It does not remove curvature and optimization errors. Effective-Hamiltonian eigenvalues alone cannot rank the branches (Evidence 1).

At $V = 0$, the separate matched $\chi = 400$ comparison gives a stripe endpoint lower than the paired endpoint by $2.4336 \times 10^{-4}t$ per physical site, or $0.03115t$ on the 128-site ladder. The September 6 working decision treated the available evidence as sufficient to regard the advantage as reasonably robust over the tested $\chi = 200$ to 400 increase. Neither original endpoint satisfies all formal acceptance gates, so the value remains a terminal diagnostic difference rather than a fully certified fixed-point ranking (Evidence 6).

The newer raw $\chi = 200$ trajectories instead end with a separation $8.99 \times 10^{-5}t$ per site while the pairing lineage is still evolving. This is not an independent estimate of the same equilibrium energy

gap. Similarly, an early low energy on an unconverged trajectory cannot be selected as a lower-energy phase. During startup, the functional combines the applied partner fields with current correlations; raw self-consistency iteration is not guaranteed to minimize a single self-consistent state's energy monotonically.

3.9 Complete square and unfrustrated-cubic grids

The September 18 review extends the anchor comparison to $t_0/t = 1.0, 1.2, 1.4$ and $V/t = 0, -0.2, -0.4$, with $U/t = 8$, $n = 0.9375$, $t_\perp/t = 0.1$, $L = 64$ rungs and $\chi = 200$. The two starts are reciprocal 95%/5% mixtures of the reference stripe and paired correlations, reconstructed with the target couplings. They evolve with raw MF updates, without Anderson acceleration. All phase descriptions below refer to finite-run physical correlations, not certified minima (Evidence 11).

After including both $(1.4, 0)$ continuations, the coarse square grid has seven stripe coordinates and two paired coordinates, $(1.4, -0.4)$ and $(1.4, -0.2)$. Both seed families now reach the same qualitative basin at each coordinate. At $(1.4, 0)$ the stripe and pairing lineages contain 100 and 82 cumulative evaluations, respectively; the late pairing collapse seen in the earlier history is therefore followed by a stripe endpoint. The whole square grid contains 902 evaluations across 18 lineages, including 20 source jobs, with no formally accepted endpoint (Evidence 12).

In contrast, all 18 unfrustrated-cubic starts reach essentially unpaired CDW/SDW stripes after 60 evaluations. Physical leg-odd spin RMS ranges from 0.280 to 0.356, while the largest leg-pair RMS is 2.21×10^{-11} . The dominant charge and spin modes are $m = 4$ and 30, corresponding to nominal 16-rung charge and 32-rung staggered-spin-envelope periods. The two square paired coordinates are striped in cubic. Every cubic endpoint passes its final energy-window, density, inner-DMRG and identity checks; remaining field/profile and slow-mode checks prevent acceptance. This is evidence of geometry-dependent nonlinear basin selection, not an energetic ranking of different Hamiltonians or an exclusion of every other cubic basin.

3.10 Finer square cuts and the shape of the variational energy

The finer cuts add $V/t = -0.15, -0.10, -0.05$ at $t_0/t = 1.4$ and $t_0/t = 1.25, 1.30, 1.35$ at $V/t = -0.4$, with 60 raw evaluations per seed. Both seeds reach paired trajectories at four new points. Distinct stripe and paired trajectories survive at $(1.25, -0.4)$ and $(1.4, -0.05)$. Their signed endpoint energy differences, pairing seed minus stripe seed, are -3.189×10^{-5} and $-5.952 \times 10^{-5} t/\text{site}$. These exceed the sums of the corresponding final-ten energy ranges, 1.574×10^{-7} and $7.461 \times 10^{-8} t/\text{site}$, but recent drift does not bound the remaining convergence error. All twelve endpoints remain unaccepted (Evidence 13).

Figure 3 shows the full stored variational energy along both cuts, including the coarse endpoints and the latest $V = 0$ continuations. The quantity is the canonical functional with its field-dependent double-counting terms and target-density correction,

$$e_{\text{target}} = E_{\text{var}}/N_s + \mu(n_{\text{target}} - n), \quad N_s = 2L = 128. \quad (9)$$

It is not the effective-Hamiltonian eigenvalue. No additional field-independent perturbative offset has been introduced. On this scale the two seed curves nearly overlap: the V cut is almost linear, while the t_0 cut has gradual curvature. Curves identify the initial seed, not a persistent phase, because independent runs can change their outcome as the parameter changes.

To resolve small shape changes, Fig. 4 subtracts one common linear chord from both seed curves on each cut. Its endpoints are the mean seed energies at the two outer coordinates. This preserves

Preliminary cubic-unfrustrated phase diagram

$L = 64, \chi = 200, n = 0.9375, t_{\perp}/t = 0.1$ | reciprocal 95% / 5% seeds

S: stripe CDW / SDW; negligible pairing



Both square paired coordinates also reach stripes here: (1.4,-0.4) and (1.4,-0.2).

All 18 endpoints remain formally unaccepted; these are observed basin assignments.

60 raw evaluations per start. No Anderson, interpolated boundary or energetic winner.

Figure 2: Unfrustrated-cubic grid from the two reference seed families. Both starts reach stripe-like endpoints at every tested coordinate. These are observed finite-run outcomes at $\chi = 200$; none is an accepted fixed point. Source: [Evidence 11](#).

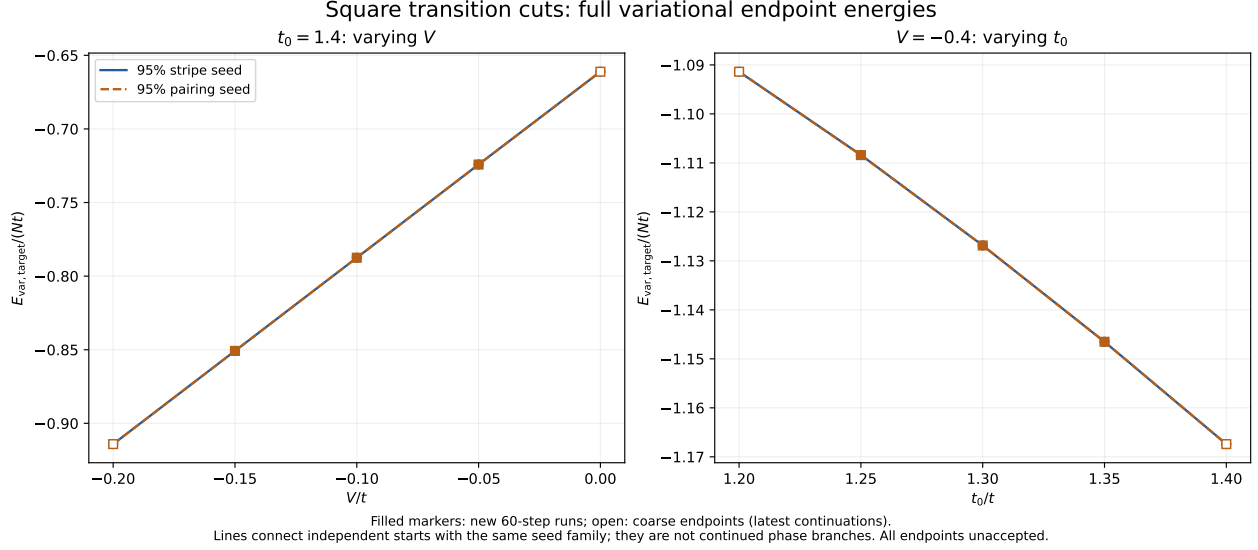


Figure 3: Full canonical variational endpoint energies, corrected to the target density. Filled points are the new 60-step starts; open points use the coarse endpoints and latest continuations. Lines guide the eye between independent starts; they are not parameter-continuation branches. Total energy equals the displayed energy per site times 128.

both seed gaps and changes in slope. The lower panels display adjacent secants $\Delta e / \Delta p$, with $p = V$ or t_0 , rather than a fitted derivative. Along the pairing-seed V curve the slopes are approximately 1.266231, 1.266144, 1.266212 and 1.261367. Thus the last interval, $-0.05 \leq V/t \leq 0$, has a slope smaller by about 0.38% than its predecessor. The stripe-seed curve also bends downward in that interval. This small feature lies near the change from paired to stripe outcomes and is compatible with a weak first-order crossing, but does not resolve one.

For a first-order transition the minimizing energy can remain continuous while its first derivative jumps. Five sampled coordinates do not distinguish that kink from a continuous change of curvature. On the t_0 cut the pairing-seed secants, -0.340515 , -0.368857 , -0.393241 , -0.417749 , change nearly regularly, with no obvious sharp cusp at this resolution. The broad arch after linear subtraction is not itself a singularity. Absence of a visible kink does not exclude a weak first-order transition.

The two distinct textures at one point on each cut make metastability and first order worth testing. However, these are not two stationary phase branches traced through a crossing, nor parameter-sweep hysteresis. The signed E_p values at the new points were linearly interpolated between the coarse endpoints as requested; $g = t_{\perp}^2 / |E_p|$ therefore also changes along each cut. The plotted slopes include this coupling dependence and cannot be identified with the bare V or rung-hopping conjugate observable alone. Coarse and fine runs also have different stopping controls.

The weak spin at the paired $(1.4, -0.05)$ endpoint requires a spatial qualification. Its usual spin RMS on rungs 6–59 rises 1.50% over the last ten evaluations to 0.005606, but 96.7% of its spin-squared weight lies in the outer 14 rungs at each end. Central 32-rung spin still decreases, while leg-pair RMS remains 0.032991. This is not established bulk coexistence or a demonstrated bulk return to stripes. At paired $(1.25, -0.4)$, the spin remainder decreases by 52.1% over the final ten evaluations.

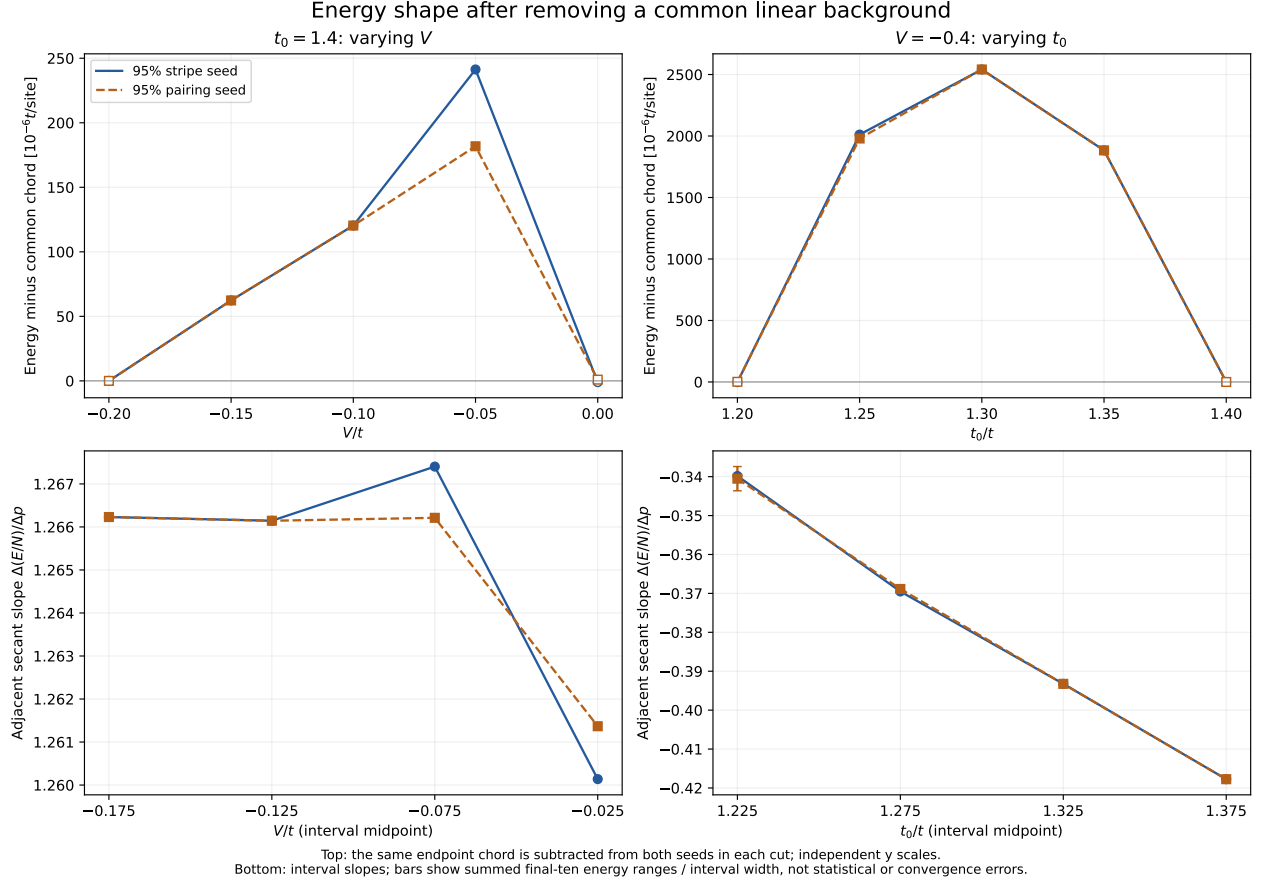


Figure 4: Energy shape and adjacent-interval slopes. The same chord is subtracted from both seed curves within each cut; the two cuts have independent vertical scales. Bars on the secants are the sum of endpoint final-ten energy ranges divided by interval width, not statistical or convergence error bounds. In particular, the coarse $t_0 = 1.2$ pairing lineage still has an energy range $1.56 \times 10^{-4} t/\text{site}$. Polygonal corners between samples are not measured derivative jumps.

3.11 Positive interaction and the first trellis result

The square $(t_0/t, V/t) = (1.2, +0.2)$ test has three completed 60-step starts: stripe, paired, and an intertwined period-16 pattern. All lose their anomalous pairing, reaching spin RMS near 0.233 and leg-pair RMS below 8.3×10^{-10} , with nominal charge/spin-envelope periods 16/32. The period-eight intertwined start has 46 complete scalar log records but no synchronized spatial state. The three complete results do not reproduce the legacy frustrated-cubic coexistence and do not support a geometry-independent rule that repulsive V promotes intertwining. The missing wavelength test and other possible basins remain open (Evidence 14).

The first completed trellis run gives the opposite conversion. At $(t_0/t, V/t) = (1, 0)$ and zigzag hoppings $\tau_0/t = \tau_1/t = 0.1$, the one-ladder stripe-dominated start develops pairing over 60 cell sweeps. Physical spin RMS decreases from 0.0627 to 1.21×10^{-5} and leg-pair RMS increases from 0.00965 to 0.01797. The final mean leg/rung amplitudes have opposite signs, $+0.01796/-0.03292$. Square and unfrustrated cubic instead reach stripes at the same bare-ladder coordinate. This comparison uses physical correlations; trellis fields include additional normal-bond Hartree contributions and cannot be inverted with the square formula.

The trellis global residual checks pass late, but energy, inner-DMRG and channel/profile gates still fail. The last-ten energy range is $2.84 \times 10^{-7} t/\text{site}$. Other synchronized logs contain 21 one-ladder pairing, 18 two-ladder stripe and one two-ladder pairing cell sweeps, without spatial artifacts. The skew one-ladder repetition and rectangular two-ladder A/B cell are distinct spatial ansatzes, not temporal two-cycle phases. No comparison of completed trellis branches or cross-geometry energies is yet justified (Evidence 15).

3.12 Scope of the emerging physical interpretation

The isolated-ladder and coupled-array observations separate pair formation from ordered-state selection. A spin-gapped ladder with slowly decaying pair correlations can still feed a static stripe state once transverse normal fields are allowed to develop. At the attractive anchor, both tested starts instead approach the same pairing-dominated solution. The current evidence therefore supports a model-specific competition between collective orders within a system that already possesses local pairing tendencies.

This result shares an ingredient with stripe-centered pairing mechanisms: correlated low-dimensional subsystems can support pairing before collective coherence is established. At the level of the current static textures, however, the anchor comparison resembles competition between stripe order and superconducting order. It neither establishes that fluctuating stripes are necessary for pairing nor excludes their possible beneficial role. The approximation treats quantum fluctuations within the ladder, but does not resolve the full collective stripe and phase fluctuations of a plane.

The most discriminating open physical question is whether the eventual stripe state retains strong pair-pair correlations after its anomalous expectation value has vanished. Strong residual pair correlations would support a distinction between local pairing and interladder coherence. Suppression of those correlations would instead indicate more direct competition with pairing. Resolving this requires measurements in the converged stripe state; the current source-field history alone does not answer it.

Attraction changes more than one ingredient of the coupled problem. The prefactor $g = t_\perp^2/|E_p|$ decreases when $|E_p|$ increases at fixed $t_\perp$, even while the ladder's pair response can become stronger. This motivates separating the change in the bare ladder from the change in the coupling kernel

across a V scan. Recent MPS+MF work on a different, mixed-dimensional t - J model likewise emphasizes the properties of the one-dimensional building blocks and the requirements for an ordered array [34]. That work is retained as an August 2026 preprint, not a result for the present Hamiltonian.

3.13 Controls needed before stronger claims

The pair-breaking expansion needs a careful energy-scale comparison. The backbone gives $t_{\perp}/|E_p| \simeq 0.683$ and $t_{\perp}/\Delta_s \simeq 0.652$ at $t_{\perp}/t = 0.1$; these ratios are not parametrically small. The quantity

$$\Delta_{c,2} = \frac{E(N+2) + E(N-2) - 2E(N)}{2} \quad (10)$$

instead measures even-particle charge curvature. A gapless charge mode can coexist with a finite spin or pair-breaking scale, so $t_{\perp}/\Delta_{c,2} > 1$ does not by itself invalidate the projection. This corrects the stronger interpretation in the earliest bare-ladder report (Evidence 5). The registry and backbone also use slightly different binding energies, differing by about 0.0188% in the inferred coupling; quantitative response/SCF comparisons should declare a common kernel.

The remaining high-value controls are:

1. Complete self-consistency at the representative stripe point and compare matched canonical energies only where the states and provenance support it. A converged fixed point and a global minimum remain different claims.
2. Test selected lengths, bond dimensions, and stripe wavelengths. The $L = 96$ and 128 seeds are prepared, but no result from them is used here. Their planned fixed- $L = 64$ binding denominator isolates one finite-size question; it is not a new measurement of the larger ladder's binding (Evidence 16).
3. Check sensitivity to the mean-field range cutoff, currently $r_{\text{range}} = 4$ in the representative comparisons, and to weaker transverse hopping. These are model/approximation controls, separate from inner-DMRG convergence.
4. Complete amplitude and basis checks for the response pilot and measure response about a self-consistent normal reference at fixed density.
5. Separate connected fluctuations from disconnected order contributions. Subtracting $\langle O_i \rangle \langle O_j \rangle$ removes the elastic mean-field order contribution; both parts are needed to interpret structure factors. Compare pair correlations across bond classes and bulk reference positions.

The completed square and unfrustrated-cubic grids now supply contrasting finite-run phase patterns. The finer cuts identify competing textures but do not resolve the order of the transition. The September 18 campaigns contain 34 completed 60-step histories, or 2040 evaluations, all unaccepted; the earlier coarse square grid is a separate data set. No universal effect of V , certified ground-state phase diagram, transition temperature, phase stiffness, pair-density-wave state, or dynamical magnetic spectrum is claimed. The present contribution is a developing account of how local pairing tendencies and competing static orders relate in a specified coupled-ladder construction.

A Local evidence and drafting notes

A.1 Sources for project-specific statements

The following sources are repository documents and analysis artifacts, not external publications. Paths are relative to `ladder_mps_mft/docs`. The linked reports retain the underlying histories, definitions, and provenance checks. Their dated conclusions must be read together: later history analyses supersede the early suggestion of a unique paired $V = 0$ basin.

1. **Model, fields, and energy.** [Living methods notes](#), [variational functional](#), and [architecture](#). `METHODS_NOTES.tex`; `VARIATIONAL_FUNCTIONAL.md`; `ARCHITECTURE.md`.
2. **Literature and equation-level geometry correspondence.** [Literature review](#) and [source notes](#), including the September 8 comparison with the 2023 benchmark. `literature/literature_review.tex`; `literature/SOURCE_NOTES.md`.
3. **Isolated-ladder backbone and covariance survey, September 2.** [Stage 1 report](#). `reports/bare_stage1_t014_v0_20260902/REPORT.md`.
4. **Projected response discovery, September 3.** [Stage 2 report](#), with eigenvalues and reference-field summaries. `reports/bare_stage2_t014_v0_20260903/`.
5. **Cross-project scientific review, September 4.** [Review](#). Source for the response limitations, charge-gap correction, and proposed approximation controls; its campaign statuses are historical. `reports/systematic_review_20260904/REVIEW.md`.
6. **Bond-dimension comparison, September 5–6.** [Analysis](#), including the explicitly qualified September 6 energetic interpretation. `reports/chi400_comparison_20260905/ANALYSIS.md`.
7. **History-based revision of basin interpretation, September 8.** [Basin assessment](#). `reports/square_grid_20260908/BASIN_ASSESSMENT.md`.
8. **Square coverage and the flagged $(1.0, -0.4)$ endpoint, September 8.** [Grid description](#) and [divergence analysis](#). `reports/square_grid_20260908/`.
9. **Attractive two-family anchors, September 10.** [Analysis](#); complementary retrospective gate qualification in the [remainder contract](#). `reports/two_basin_vm04_20260910/README.md`.
10. **Zero-attraction anchors and slow texture relaxation, September 12–13.** [Analysis and follow-up](#). Figure 1 is reused without modifying the underlying data. `reports/two_basin_v000_20260912/README.md`.
11. **Combined synchronized campaign review, September 18.** [Combined report](#) and [complete cubic grid](#). Source inventories, physical profiles, gate checks and accounting coverage are retained with the detailed reports.
12. **Complete coarse square grid and continuations.** [Updated square report](#), including both $(1.4, 0)$ continuations and the historical parent records.

13. **Finer square cuts and full energy curves, September 18.** [Cut report](#), [endpoint energy table](#), and [energy audit](#). Twenty endpoint sources include the twelve fine runs and eight coarse anchors.
14. **Square repulsive-interaction test, September 18.** [Four-seed comparison](#); three spatial endpoints and one partial scalar log.
15. **First trellis endpoint and partial logs, September 18.** [Trellis report](#) and [spatial-cell method contract](#).
16. **Prepared finite-size seeds, September 6.** [Seed description](#). Preparation evidence only; no larger-length simulation result is inferred. `reports/finite_size_seeds_20260906/REA DME.md`.

A.2 Reference provenance and manuscript development

The draft reuses stable keys from `../literature/references.bib`. Five additional sources are kept in `additional_references.bib`: Tranquada et al. (2004), Vojta and Ulbricht (2004), Li et al. (2007), Berg et al. (2007), and Missiaen et al. (2025). Their primary arXiv and publisher records were checked on 15 September 2026. These additions support the discussion of magnetic spectra, planar coherence, and the stripe–pseudogap connection. The existing review’s reading-depth limits continue to apply to inherited references; this draft is not an exhaustive literature or novelty assessment.

The introduction can be shortened for a target journal after the principal result and convergence controls are settled. The current results collection should retain its date and evidence qualifications until replaced by converged comparisons. A future manuscript should decide whether its central claim is interaction-driven order selection, geometry dependence, or the relation between a dressed normal-state response and the nonlinear ordered solutions. Those are candidate emphases, not additional results.

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